

```
$ sudo service postgresql start
```

You can now install the *postgresql-simple* package using the *cabal* command:

```
$ cabal install postgresql-simple
```

Let us first create a database and a schema using the PostgreSQL command-line utility *psql*:

```
$ psql -U postgres
Password for user postgres:
psql (9.3.5)
Type "help" for help.

postgres=# \l
                                List of databases
Name | Owner  | Encoding | Collate | Ctype  | Access privileges
-----+-----+-----+-----+-----+-----
postgres | postgres | UTF8    | en_US.UTF-8 | en_US.UTF-8 |
template0 | postgres | UTF8    | en_US.UTF-8 | en_US.UTF-8 |
=c/ postgres |
postgres=Ctc/postgres
template1 | postgres | UTF8    | en_US.UTF-8 | en_US.UTF-8 |
=c/postgres |
postgres=Ctc/postgres
(3 rows)

postgres=# CREATE DATABASE test;
CREATE DATABASE

postgres=# \c test
```

You are now connected to database 'test' as user 'postgres'.

```
test=# create schema social;
CREATE SCHEMA

test=# \dn
      public | postgres
      social | postgres
```

We can then create a users' table with an ID, first name and last name using the *postgresql-simple* package:

```
{-# LANGUAGE OverloadedStrings #-}

import Database.PostgreSQL.Simple

main :: IO ()
main = do
  conn <- connect defaultConnectInfo
    { connectUser = "postgres"
```

```
    , connectPassword = "postgres123"
    , connectDatabase = "test"
  }
```

```
  execute conn "create table social.users (id INT, fname
VARCHAR(80), lname VARCHAR(80))" ()
  close conn
```

Loading the above code in GHCi creates the table *social.users* as shown below:

```
$ ghci create.hs
```

```
GHCi, version 7.6.3: http://www.haskell.org/ghc/ :? for help
Loading package ghc-prim ... linking ... done.
Loading package integer-gmp ... linking ... done.
Loading package base ... linking ... done.
[1 of 1] Compiling Main             ( create.hs, interpreted )
Ok, modules loaded: Main.
```

```
ghci> main
```

```
Loading package array-0.4.0.1 ... linking ... done.
Loading package deepseq-1.3.0.1 ... linking ... done.
Loading package bytestring-0.10.0.2 ... linking ... done.
Loading package containers-0.5.0.0 ... linking ... done.
Loading package text-0.11.3.1 ... linking ... done.
Loading package attoparsec-0.10.4.0 ... linking ... done.
Loading package blaze-builder-0.3.1.1 ... linking ... done.
Loading package dlist-0.5 ... linking ... done.
Loading package hashable-1.1.2.5 ... linking ... done.
Loading package transformers-0.3.0.0 ... linking ... done.
Loading package mtl-2.1.2 ... linking ... done.
Loading package old-locale-1.0.0.5 ... linking ... done.
Loading package syb-0.4.0 ... linking ... done.
Loading package pretty-1.1.1.0 ... linking ... done.
Loading package template-haskell ... linking ... done.
Loading package time-1.4.0.1 ... linking ... done.
Loading package unordered-containers-0.2.3.0 ... linking ... done.
Loading package primitive-0.5.0.1 ... linking ... done.
Loading package vector-0.10.0.1 ... linking ... done.
Loading package aeson-0.6.2.1 ... linking ... done.
Loading package random-1.0.1.1 ... linking ... done.
Loading package scientific-0.2.0.2 ... linking ... done.
Loading package case-insensitive-1.0.0.1 ... linking ... done.
Loading package blaze-textual-0.2.0.8 ... linking ... done.
Loading package postgresql-libpq-0.9.0.2 ... linking ... done.
Loading package binary-0.7.4.0 ... linking ... done.
Loading package cereal-0.3.5.2 ... linking ... done.
Loading package entropy-0.2.2.1 ... linking ... done.
Loading package tagged-0.6 ... linking ... done.
Loading package crypto-api-0.11 ... linking ... done.
Loading package cryptohash-0.9.0 ... linking ... done.
Loading package network-info-0.2.0.5 ... linking ... done.
```